MIL-C-915/41A
1 August 1972
SUPERSEDING[1/]
MIL-C-915/41(SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

(A) CABLE, ELECTRICAL, 600 VOLTS, TYPE 2AU

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, ASTM B-286 size 22-7. Maximum diameter 0.033 inch.
(A) Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Minimum thickness 0.020 inch. Telephone Identification Code, applied by method 3.
(A) Third - Insulation jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.004 inch.
Fourth - Two conductors cabled together to form a pair, with nominal lay of 1-1/2 inches.
Fifth - Forty pairs cabled together, with a lay not greater than 16 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair No. 1, from the center outward.
Sixth - Binder tape, applied helically with overlap. (see note 1).
Seventh - Braided shield of AWG No. 34 tin coated copper. Minimum coverage of 85 percent.
Eighth - Binder tape applied helically with overlap.
(A) Ninth - Standard thermoplastic jacket (gray). Nominal thickness 0.060 inch. Cable surface marking required.

(A)

Details

Type and size	Number of pairs	Overall diameter	
		Min. inches	Max. inches
2AU-40	40	1.320	1.370

NOTES:
1. Since the capacitance of all pairs should be approximately the same, it may be necessary to apply a greater thickness of insulation to conductors in the outer layer, because of their proximity to the overall shield braid. For the same reason, the first binder tape should be one having good electrical properties.

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
(A) Conductor resistance - At 25°C., ohms/1000 ft., max.	17.71
Voltage withstand - Volts, rms, minimum,	
Conductor to shield	2500
Shield to water	(A) 1000
Insulation resistance - Megohms/1000 ft., minimum	
Conductor to conductor	5000

[1/] Requirements formerly covered on MIL-C-24145/1.

(A) Denotes changes.

FSC 6145

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	Requirements:
Group A:	
Visual and dimensional	
Capacitance - At 1 MHz, pF/ft., maximum	
Grounded capacitance	20
Mutual capacitance	20
Group B:	
(A) Cold bending, cable - At 0° ± 2°C., mandrel diameter 12 times cable diameter	No damage
(A) Bending endurance - At 25° ± 2°C., cycles, minimum	25
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (jacket)	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTH: 1000 feet (nominal)

(A) Custodians:
Army - MI
Navy - SH
Air Force - 80

(A) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(A) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0609-29)

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